# CLOUD COMPUTING

Multiple-Choice Questions • Questions, Options & Answer Key • 10 Chapters

**Study note:** Questions are transcribed and lightly cleaned up from the supplied images. A few source questions have ambiguous wording or potentially disputed textbook answers; these are flagged in the notes at the end.

## Chapter 1 — Capacity Planning in Cloud Computing

**1. The concept of delivering infinite resources in cloud computing is achieved by:**

A) Maintaining a huge pool of resources
B) Workload management
C) Proper capacity planning
D) None of these

**Correct answer: A — Maintaining a huge pool of resources**

**2. Over-provisioning of resources in traditional capacity planning causes:**

A) Performance degradation
B) Loss of reputation
C) Performance enhancement
D) Wastage of budget

**Correct answer: D — Wastage of budget**

**3. Under-provisioning of resources in traditional capacity planning may cause:**

A) Auto-scaling
B) Loss of reputation
C) Performance enhancement
D) Budget saving

**Correct answer: B — Loss of reputation**

**4. The availability of infinite resources on demand in cloud computing is actually an illusion.**

A) True
B) False

**Correct answer: A — True**

**5. Unlike the N + 1 rule of capacity maintenance, cloud service consumers enjoy which facility?**

A) N + X rule
B) N rule
C) Infinite resource
D) Auto-scaling

**Correct answer: C — Infinite resource**

**6. A resource utilization curve represents:**

A) Average resource utilization
B) Total volume of system resources
C) Resource utilization under load
D) All of these

**Correct answer: C — Resource utilization under load**

**7. When resource utilization reaches its ceiling, that limit is called:**

A) Stress point
B) Limiting point
C) End point
D) None of these

**Correct answer: A — Stress point**

**8. Unlike traditional computing, capacity planning is no longer required to avail computing facilities in the cloud.**

A) True
B) Required, but much easier
C) Required, but less risky
D) Both B and C

**Correct answer: D — Both B and C**

**9. Before capacity enhancement of an application, the capacity planner should know about:**

A) Cost of the additional resources
B) Value of the added capacity in business
C) How much resource is required
D) Both A and C

**Correct answer: D — Both A and C**

**10. IaaS consumers in the cloud do capacity planning with:**

A) Virtual resources
B) Physical resources
C) Both physical and virtual resources
D) None of these

**Correct answer: A — Virtual resources**

**11. What is the relation between the capacity of a system and its performance?**

A) Proportional
B) Reverse
C) Similar
D) None of these

**Correct answer: A — Proportional**

## Chapter 2 — Cloud Computing Fundamentals

**1. Which of the following is not a limitation of the traditional computing approach?**

A) Huge initial investment
B) Hardware-system management responsibility
C) Partial utilization of resources
D) None of these

**Correct answer: D — None of these**

**2. Which of the following is an advantage of the cloud computing approach?**

A) It is ubiquitous
B) Enabled by pay-for-usage model
C) Responsiveness
D) All of these

**Correct answer: D — All of these**

**3. Access to a cloud environment costs more money compared to a traditional environment.**

A) True
B) False

**Correct answer: B — False**

**4. A cloud environment can be accessed from anywhere in the world as long as the user has access to the Internet.**

A) True
B) False

**Correct answer: A — True**

**5. Which one of the following is the standard cloud computing pricing model?**

A) Free
B) Pay-per-usage
C) Licensing basis
D) None of these

**Correct answer: B — Pay-per-usage**

**6. Cloud computing makes a business more:**

A) Dynamic
B) Strong
C) Profitable
D) Simple

**Correct answer: A — Dynamic**

**7. Which among the following is a feature of utility service?**

A) 24×7 availability
B) Unlimited access
C) Metering
D) None of these

**Correct answer: C — Metering**

**8. A computing system is composed of three facility layers. This statement is:**

A) True
B) False

**Correct answer: A — True**

**9. Which among the following is not considered as a layer in a computing system?**

A) Application
B) Server
C) Infrastructure
D) Platform

**Correct answer: B — Server**

**10. Computing network facility falls under which layer of computing?**

A) Application
B) Platform
C) Infrastructure
D) Both B and C

**Correct answer: C — Infrastructure**

**11. Is the IT outsourcing model of traditional computing similar to cloud computing?**

A) Yes
B) Not at all
C) To some extent
D) Depends on the requirement

**Correct answer: C — To some extent**

**12. What is the main advantage of utility computing compared to traditional computing?**

A) Less difficulty
B) Better security
C) Economic
D) Availability

**Correct answer: C — Economic**

## Chapter 3 — History and Evolution of Cloud Computing

**1. The idea of cloud computing is entirely new and surfaced at the beginning of the current century.**

A) True
B) False, the idea was there from long back
C) False, it didn't materialize earlier due to technical constraints
D) Both B and C

**Correct answer: D — Both B and C**

**2. The concept of computing as a public utility was first presented by John McCarthy in which year?**

A) 1960
B) 1970
C) 1961
D) 2000

**Correct answer: C — 1961**

**3. Which of the following is an important facilitator for the emergence of distributed computing?**

A) The invention of personal computers
B) Emergence of distributed applications
C) Fast network communication
D) None of these

**Correct answer: C — Fast network communication**

**4. With which model of distributed computing did the idea of resource pool appear?**

A) Cluster computing
B) Grid computing
C) Client-server computing
D) None of these

**Correct answer: B — Grid computing**

**5. Which of the following is/are advantage(s) of grid computing over cluster model?**

A) Solution of the cluster head problem
B) Heterogeneous resource pool
C) Co-operation between multiple administrative domains
D) All of these

**Correct answer: D — All of these**

**6. The difficulty of maintaining mobility of an application over a heterogeneous resource pool was solved with:**

A) Uniform system management policy
B) Resource virtualization
C) Clustering
D) None of these

**Correct answer: B — Resource virtualization**

**7. Web-based system development has benefited with the emergence of:**

A) Web 2.0
B) Mashup
C) SOA
D) All of these

**Correct answer: D — All of these**

**8. Autonomic computing is a means for making the management of computing infrastructure:**

A) Automatic and faster
B) Automatic and intelligent
C) Automatic and cheaper
D) None of these

**Correct answer: B — Automatic and intelligent**

**9. Cloud computing is the result of the convergence of:**

A) Grid computing and hardware virtualization
B) Web technology and SOA
C) High-speed network and utility model
D) All of these
**Correct answer: D — All of these**

## Chapter 4 — Utility Computing and Web Services

**1. The utility model of computing is beneficial for:**

A) Users of computing
B) Vendors of computing
C) Both B and C
D) None
**Correct answer: C — Both users and vendors of computing (the option wording in the image appears to say "Both b and c"; likely intended both users and vendors)**

**2. The term "cloud computing" was first presented in its current meaning in the year:**

A) 1999
B) 2006
C) 2008
D) 2010
**Correct answer: B — 2006**

**3. Which among the following are feature(s) of utility service model?**

A) Metering
B) Billing
C) On-demand service availability
D) All of these
**Correct answer: D — All of these**

**4. Which among the following companies was first to successfully adopt utility computing model for enterprises?**

A) Amazon
B) Salesforce.com
C) IBM
D) Google
**Correct answer: C — IBM**

**5. Heterogeneous software applications communicate over the Internet using:**

A) HTML
B) Message passing
C) Web services
D) None of these
**Correct answer: C — Web services**

**6. Presently, web services are categorized as:**

A) SOAP & REST
B) SOAP & RPC
C) REST & RMI
D) REST & CORBA
**Correct answer: A — SOAP & REST**

**7. REST is considered as the light-weight alternative of SOAP since it can avoid:**

A) Use of global identifier for resources
B) Use of XML for message passing
C) Use of XML for message passing
D) None of these

**Correct answer: B — Use of XML for message passing**

**8. Use of REST-compliant web services results in longer development time than using SOAP-based web services in cloud computing development. This statement is:**

A) True
B) False

**Correct answer: B — False**

**9. REST is a/an:**

A) Message communication protocol
B) Architectural style
C) Web architecture
D) Cloud type

**Correct answer: B — Architectural style**

**10. Which among the following is not a benefit of cloud computing?**

A) Low initial investment
B) Fixed cost of computing
C) Access through Internet
D) High-performance computing

**Correct answer: B — Fixed cost of computing**

**11. Which among the following is not considered an inherent nature of cloud computing?**

A) Flexibility
B) Elasticity
C) Scalability
D) Portability

**Correct answer: D — Portability**

## Chapter 5 — NIST Cloud Model and Deployment

**1. Which of the following is not an essential characteristic of the NIST cloud model?**

A) Rapid elasticity
B) Multi-tenancy
C) Resource pooling
D) Broad network access

**Correct answer: B — Multi-tenancy**

**2. The NIST definition of cloud computing was published in the year:**

A) 2007
B) 2009
C) 2011
D) 2013

**Correct answer: B — 2009**

**3. According to the NIST definition, cloud computing is a:**

A) Model
B) Technology
C) Paradigm
D) Revolution

**Correct answer: A — Model**

**4. Which among the following service can only be accessed by a restricted number of people?**

A) Public cloud
B) Community cloud
C) Private cloud
D) Both A and B

**Correct answer: C — Private cloud**

**5. Which among the essential characteristics of cloud computing is not present in the grid computing model?**

A) Rapid elasticity
B) On-demand service
C) Measured service
D) Both A and B

**Correct answer: D — Both A and B**

**6. Public cloud can't be deployed on-premises.**

A) False
B) True

**Correct answer: B — True**

**7. Private cloud can be deployed on-premises. This statement is:**

A) False
B) True

**Correct answer: B — True**

**8. Users' control over cloud environment is maximum in:**

A) Public cloud
B) Community cloud
C) Private cloud
D) Hybrid cloud

**Correct answer: C — Private cloud**

**9. Hybrid cloud is cheaper than private cloud. This statement is:**

A) True
B) False

**Correct answer: B — False (not universally true; cost depends on workload and configuration)**

**10. Which among the following refers to the location and management of the cloud's infrastructure?**

A) Application
B) Services
C) Deployment
D) All of these

**Correct answer: C — Deployment**

**11. Which cloud deployment is managed by provider who resides offsite and is accessible to some restricted people?**

A) Private cloud
B) Community cloud
C) Public cloud
D) Hybrid cloud

**Correct answer: B — Community cloud**

**12. The number of actors mentioned in the NIST cloud computing reference architecture is:**

A) 4
B) 5
C) 6
D) 7

**Correct answer: B — 5**

# Chapter 6 — Cloud Service Models

**1. Which cloud service is considered as the most widely used?**

A) Infrastructure-as-a-Service
B) Platform-as-a-Service
C) Software-as-a-Service
D) All of these

**Correct answer: C — Software-as-a-Service**

**2. Which cloud service is known as Hardware-as-a-Service also?**

A) Infrastructure-as-a-Service
B) Platform-as-a-Service
C) Software-as-a-Service
D) Desktop-as-a-Service

**Correct answer: A — Infrastructure-as-a-Service**

**3. Which of the following describes a distribution model in which applications are hosted by a service provider and made available to users?**

A) Infrastructure-as-a-Service
B) Platform-as-a-Service
C) Software-as-a-Service
D) All of these

**Correct answer: C — Software-as-a-Service**

**4. Virtual machine in Rackspace cloud is called:**

A) Rackspace Server
B) Cloud Server
C) RackVM
D) None of these

**Correct answer: B — Cloud Server**

**5. Identify the odd one out:**

A) Microsoft Azure
B) Google App Engine
C) Force.com
D) Amazon EC2

**Correct answer: D — Amazon EC2 (the first three are commonly associated with PaaS; EC2 is IaaS)**

**6. In IaaS, computing resources are delivered to consumers:**

A) Through remote access
B) Through distribution
C) In virtualized form
D) None of these

**Correct answer: C — In virtualized form**

**7. Which is not considered as one of the three main categories of cloud services?**

A) Software-as-a-Service
B) Database-as-a-Service
C) Platform-as-a-Service
D) Infrastructure-as-a-Service

**Correct answer: B — Database-as-a-Service**

**8. Which cloud computing service model delivers computing, storage, and network to consumers?**

A) Infrastructure-as-a-Service
B) Platform-as-a-Service
C) Software-as-a-Service
D) All of these

**Correct answer: A — Infrastructure-as-a-Service**

**9. Retaining a traditional computing environment costs more compared to using cloud computing.**

A) True
B) False

**Correct answer: A — True (as a general textbook comparison; actual costs vary)**

**10. Name of the open-source cloud computing PaaS facility originally developed by VMware is:**

A) VMware platform
B) Cloud Foundry
C) VMW-PaaS
D) None of these

**Correct answer: B — Cloud Foundry**

**11. Eucalyptus, Nebula, Nimbus, CloudStack, VCL are examples of:**

A) Open-source cloud services
B) Closed-source cloud services
C) Proprietary cloud services
D) Free cloud services

**Correct answer: A — Open-source cloud services**

## Chapter 7 — Cloud Security

**1. What is the name of the organization majorly helping to foster security standards for cloud computing?**

A) NIST
B) Cloud Security WatchDog
C) Cloud Security Alliance
D) Open Cloud Consortium

**Correct answer: A — NIST**

**2. “Perimeter” in the perimetered computing environment is represented by:**

A) Organization’s physical boundary
B) In-house data center boundary
C) IP address
D) Network firewalls

**Correct answer: A — Organization’s physical boundary**

**3. On how many elements of cloud security model was it focused in Gartner’s report titled “Assessing the Security Risks of Cloud Computing” published in 2008?**

A) 6
B) 7
C) 8
D) None of these

**Correct answer: B — 7**

**4. Who proposed the globally-accepted model for cloud security?**

A) Cloud Security Alliance
B) Jericho Forum group
C) NIST
D) Object Management Group (OMG)

**Correct answer: B — Jericho Forum group**

**5. Into how many visible cubes was the model cube divided in Jericho Forum’s security model?**

A) 8
B) 16
C) 4
D) 32

**Correct answer: A — 8**

**6. How many dimensions exist in Cloud Cube Model?**

A) 1
B) 2
C) 3
D) 4

**Correct answer: D — 4**

**7. Which among the following security policies demands special attention in cloud computing?**

A) Data security policy
B) Security management policy
C) Advisory policy
D) Regulatory policy

**Correct answer: A — Data security policy (likely intended based on the listed choices)**

**8. Which among the four dimensions of the Cloud Cube Model is represented by the colour?**

A) Data Boundary
B) Ownership
C) Security Boundary
D) Sourcing

**Correct answer: D — Sourcing**

**9. De-perimeterization in computing promotes:**

A) Outsourcing
B) Use of open technology
C) Collaboration
D) All of these

**Correct answer: D — All of these**

**10. The statement "internal data source can be used in de-perimeterized computing environment" is:**

A) True
B) False

**Correct answer: A — True**

**11. In traditional perimetered computing environment, collaboration with external systems is established through:**

A) Virtual private network technique
B) Virtual drive technique
C) De-perimeterization technique
D) None of these

**Correct answer: A — Virtual private network technique**

## Chapter 8 — Virtualization

**1. Which of these attributes in a system could be brought by computing infrastructure virtualization?**

A) Security
B) Dynamic behaviour
C) Flexibility
D) All of these

**Correct answer: D — All of these**

**2. The physical system on which virtual machines run is called:**

A) Host machine
B) Guest machine
C) Primary machine
D) Non-virtual machine

**Correct answer: A — Host machine**

**3. The software layer implementing virtualization is known as:**

A) Operating system
B) Hypervisor
C) Application layer
D) None of these

**Correct answer: B — Hypervisor**

**4. In server virtualization, the relationship between host machine and guest machine is:**

A) Many-to-one
B) One-to-one
C) Many-to-many
D) One-to-many

**Correct answer: D — One-to-many**

**5. Is it possible to install an operating system in a virtual machine?**

A) Yes
B) No
C) Sometimes
D) OS is not required in virtual machine

**Correct answer: A — Yes**

**6. Type 1 hypervisors are installed over:**

A) Physical resource
B) Host operating system
C) Virtual resource
D) Type 2 hypervisor

**Correct answer: A — Physical resource**

**7. Type 2 hypervisors are installed over:**

A) Physical resource
B) Host operating system
C) Virtual resource
D) Type 1 hypervisor

**Correct answer: B — Host operating system**

**8. Operating systems and applications made for one computer architecture can be installed on some other architecture in:**

A) Simple server virtualization
B) Simulation-based server virtualization
C) Complex server virtualization
D) Emulation-based server virtualization

**Correct answer: D — Emulation-based server virtualization**

**9. Guest operating systems remain unaware that they are running in virtual environment in:**

A) Full virtualization
B) Para-virtualization
C) Operating system-level virtualization
D) Hardware-assisted virtualization

**Correct answer: A — Full virtualization**

**10. Hypervisor-dependent (modified) versions of guest operating systems are required in:**

A) Full virtualization
B) Para-virtualization
C) Hardware-assisted virtualization
D) All of these

**Correct answer: B — Para-virtualization**

**11. Virtualization overhead of hypervisor is maximum in case of:**

A) Full virtualization
B) Para-virtualization
C) Hardware-assisted virtualization
D) Equal for all

**Correct answer: A — Full virtualization**

**12. Virtual machines running over a physical host do not have different operating systems in:**

A) Full virtualization
B) Operating system-level virtualization
C) Para-virtualization
D) Hardware-assisted virtualization

**Correct answer: B — Operating system-level virtualization**

**13. The most popular open-source hypervisor available in the market is:**

A) ESX
B) ESXi
C) Hyper-V
D) Xen

**Correct answer: D — Xen**

**14. Process virtual machines are made to run:**

A) Operating system
B) Operating system and applications
C) Some specific application
D) Any application

**Correct answer: C — Some specific application**

**15. Server virtualization creates scope for:**

A) Speeding up servers
B) Fast installation of servers
C) Server portability
D) Both B and C

**Correct answer: D — Both B and C**

**16. Resource abstraction in cloud computing is implemented through:**

A) Resource management
B) Resource virtualization
C) Object orientation
D) Remote access

**Correct answer: B — Resource virtualization**

**17. Virtual Machine Monitor (VMM) is the other name of:**

A) Guest system
B) Host system
C) Host operating system
D) Hypervisor

**Correct answer: D — Hypervisor**

**18. In operating system-level virtualization, different OS distributions of same kernel can be run in virtual servers.**

A) True
B) False

**Correct answer: A — True**

**19. Cloud computing IaaS services can be developed without implementing resource virtualization also.**

A) True
B) False

**Correct answer: B — False**

**20. Which among the following must be maintained to ensure security in virtualization?**

A) Hardening the hypervisor
B) Restrict physical access to host machine
C) Maintain one function per VM
D) All of these

**Correct answer: D — All of these**

**21. The “single point” in the “single point of failure problem” of virtualization is:**

A) Virtual machine
B) Guest OS
C) Host machine
D) VMM

**Correct answer: C — Host machine**

**22. In “VM escape”, which among the following happens?**

A) Accessing guest OS by escaping VM
B) Evading security of VM to access VMM
C) Accessing host OS by escaping VM
D) Escaping security of VMM to access host

**Correct answer: D — Escaping security of VMM to access host**

**23. Native virtualization machines may cause security threats to virtualization.**

A) True
B) False

**Correct answer: A — True**

**24. The benefits of infrastructure virtualization include:**

A) Better utilization of resources
B) Cost saving for computation
C) Easier capacity expansion
D) All of these

**Correct answer: D — All of these**

## Chapter 9 — Resource Pools and Provisioning

**1. By which kind of hardware components does cloud computing become cost-effective at the time of building resource pools?**

A) Commodity hardware
B) Specialized hardware
C) Costly hardware components
D) None of these

**Correct answer: A — Commodity hardware**

**2. Resource pools are organized in:**

A) Heap structure
B) Hierarchical structure
C) Hash structure
D) None of these

**Correct answer: B — Hierarchical structure**

**3. The traditional approach of having discrete and independent sets of resources is known as:**

A) Warehouse
B) Data center
C) Silos
D) None of these

**Correct answer: C — Silos**

**4. Resource pools at cloud data centers are created with:**

A) Specialized hardware components
B) Commodity hardware components
C) Costly hardware components
D) None of these

**Correct answer: B — Commodity hardware components**

**5. Multi-tenancy in cloud computing is powered by:**

A) Ownership of free resource sharing
B) Temporary resource allocation
C) Resource allocation in virtual mode
D) All of these

**Correct answer: C — Resource allocation in virtual mode**

**6. In cloud computing, resource provisioning to consumers is made through:**

A) Physical servers
B) Virtual private network
C) Virtual machines
D) None of these

**Correct answer: C — Virtual machines**

**7. In static provisioning of resources, resource allocation decision is taken:**

A) Before starting of application
B) Any time before and after starting of application
C) Only once
D) Both A and C

**Correct answer: A — Before starting of application**

**8. In dynamic provisioning of resources, resource allocation decision can be taken:**

A) After starting of application
B) On-demand
C) For number of times
D) All of these

**Correct answer: D — All of these**

**9. Dynamic resource provisioning approach eliminates the problem(s) of:**

A) Resource under-provisioning
B) Resource over-provisioning
C) Both A and B
D) System speed

**Correct answer: C — Both A and B**

**10. Zero-downtime system architecture is outcome of:**

A) Dynamic provisioning
B) Multi-tenancy
C) Resource sharing
D) Resource pooling

**Correct answer: A — Dynamic provisioning**

**11. Standardization of cloud environment is done over the:**

A) Physical resource pool
B) Virtualized resource pool
C) Individual physical resources
D) None of these

**Correct answer: B — Virtualized resource pool**

## Chapter 10 — Dynamic Scaling and Scalability

**1. Which of the following is an advantage of dynamic scaling?**

A) Resource capacity can be increased during run-time
B) Resource capacity can be reduced during run-time
C) No system restart is required after resource capacity change
D) All of these

**Correct answer: D — All of these**

**2. Dynamic scaling facility in cloud has been achieved based on:**

A) Resource virtualization
B) Resource sharing
C) Dynamic resource provisioning
D) All of these

**Correct answer: D — All of these**

**3. Can dynamic scaling approach be manual?**

A) Yes
B) No

**Correct answer: A — Yes**

**4. Which of the following is the most fascinating feature of dynamic scaling in cloud computing?**

A) It is prompt
B) It is automatic
C) It is reversible
D) It is infinite

**Correct answer: B — It is automatic**

**5. Predictable load patterns in cloud computing are managed through:**

A) Dynamic scaling strategy
B) Proactive scaling strategy
C) Reactive scaling strategy
D) Static scaling strategy

**Correct answer: B — Proactive scaling strategy**

**6. In cloud computing, auto-scaling boundaries are set for a scalable system to:**

A) Make the scaling effective
B) Protect resources from external attack
C) Minimize cost
D) None of these

**Correct answer: A — Make the scaling effective**

**7. Distributed computing system architecture is apt to scale:**

A) Horizontally
B) Vertically
C) Both horizontally and vertically
D) None of these

**Correct answer: A — Horizontally**

**8. Infinite scalability can be achieved through the implementation of:**

A) Reactive scaling
B) Proactive scaling
C) Horizontal scaling
D) Vertical scaling

**Correct answer: C — Horizontal scaling**

**9. Better performance of a system indicates better scalability.**

A) True
B) False

**Correct answer: B — False**

**10. Which of the following is the primary factor to limit scalability?**

A) Cost of system management
B) Resource bottleneck
C) Growing number of users
D) Cost of resource components

**Correct answer: B — Resource bottleneck**

**11. Which type of scaling is more apt for distributed computing environment like cloud?**

A) Horizontal scaling
B) Vertical scaling
C) A mixture of horizontal and vertical scaling
D) It depends on distributed computing architecture

**Correct answer: A — Horizontal scaling**

## Notes on ambiguous or textbook-dependent items

• Chapter 4, Q1: The scanned options appear to contain a typo (“Both b and c”). The likely intended correct response is that utility computing benefits both consumers and vendors.

• Chapter 5, Q9: Whether a hybrid cloud is cheaper than a private cloud is not universally true; the cost depends on the services and deployment. The answer shown follows the likely intended True/False classroom key only if the course material states this comparison.

• Chapter 7, Q8: The Cloud Cube Model’s dimensions are commonly described as Internal/External, Proprietary/Open, Perimeterized/De-perimeterized, and Insourced/Outsourced. The scan’s question asks which dimension is represented by colour; “Sourcing” is the likely intended answer in the source’s framing.

• Some question stems/options in the screenshots contain typographical errors or unusual wording. They have been normalized for readability without changing the apparent meaning.